實習生月報

PDO 資料路徑重建

CoE / SDO 交易重建

PDO Mapping / Assignment 關聯整理

SyncManager / FMMU 資料路徑分析

報告期間

2026.08

MONTHLY REPORT

范舜傑 | MDSBU 通訊工程課 | 08/31/2026

△ DELTA LOGO

本月完成摘要

01	SDO TRANSACTION	CoE / SDO transaction reconstruction
02	PDO MAPPING	Mapping / Assignment relationship reconstruction
03	SM / FMMU	Process data path analysis
04	ADDRESS BINDING	Logical region to configured slave binding

本月工作大綱

01

02

03

04

WEEK 01

CoE / SDO

SDO transaction reconstruction

WEEK 02

PDO Mapping

Mapping object reconstruction

WEEK 03

SyncManager

SM2 / SM3 configuration analysis

WEEK 04

FMMU

Logical process data path analysis

SECTION / 01

01 CoE / SDO

交易重建

從 EtherCAT Mailbox / CoE 封包還原 SDO object access

MAILBOX —— CoE —— SDO

EtherCAT Command / Addressing

01 DEVICE / POSITION

Auto Increment

利用 position 遞增語意逐站命中 slave

02 DEVICE / NODE

Configured Address

以 configured station address 定向存取單一 slave

03 DEVICE / ALL

Broadcast

對所有 slave 的相同 local offset 共同操作

04 LOGICAL / FMMU

Logical Address

以 logical address 經 FMMU 映射至 local memory

04 / LOGICAL / FMMU

Logical Address

指令 LRD / LWR / LRW

地址欄位 32-bit Logical Address

核心機制

Master 存取的是 logical address，而不是直接指定某個 slave 的 ADP / ADO。邏輯位址範圍會經由 FMMU 映射到對應 slave 的 local memory，進而形成 process data 的資料路徑。

補充說明

這個 family 的重點已從「命中哪個 slave」，轉成「logical region 如何被映射到實際資料區塊」。

封包觀察

分析時要關注：

- logical address 範圍
- 對應 FMMU 映射結果
- 最終落到哪些 slave / local memory 區段
- 是否為 PDO / cyclic process data path

典型用途

- PDO / process data 週期性讀寫
- 邏輯位址空間的 cyclic access
- 與 FMMU / SyncManager 分析串接

結論

Logical Address 是從「資料路徑與映射結果」出發的存取方式。

Startup Register Access

[REGISTER GROUPS]

[REGISTER GROUP A]

[ADDRESS]	[NAME]	[SHORT PURPOSE]
[ADDRESS]	[NAME]	[SHORT PURPOSE]
[ADDRESS]	[NAME]	[SHORT PURPOSE]

[REGISTER GROUP B]

[ADDRESS]	[NAME]	[SHORT PURPOSE]
[ADDRESS]	[NAME]	[SHORT PURPOSE]
[ADDRESS]	[NAME]	[SHORT PURPOSE]

[REGISTER GROUP C]

[ADDRESS]	[NAME]	[SHORT PURPOSE]
[ADDRESS]	[NAME]	[SHORT PURPOSE]
[ADDRESS]	[NAME]	[SHORT PURPOSE]

[REGISTER GROUP D]

[ADDRESS]	[NAME]	[SHORT PURPOSE]
[ADDRESS]	[NAME]	[SHORT PURPOSE]
[ADDRESS]	[NAME]	[SHORT PURPOSE]

[COMMON OPERATIONS]

[COMMON OPERATION]

[COMMON OPERATION]

[COMMON OPERATION]

[COMMON OPERATION]

[COMMON OPERATION]

[IMPORTANT NOTE TITLE]

[IMPORTANT NOTE BODY]

Auto Increment → Slave Discovery

[STEP 01]

[STEP TITLE]

[STEP BODY]

[STEP 02]

[STEP TITLE]

[STEP BODY]

[STEP 03]

[STEP TITLE]

[STEP BODY]

[STEP 04]

[STEP TITLE]

[STEP BODY]

[STEP 05]

[STEP TITLE]

[STEP BODY]

[RESULT TITLE]

[RESULT ITEM]	[RESULT ITEM]	[RESULT ITEM]	[RESULT ITEM]	[RESULT ITEM]

分析問題與目標

01 PROBLEM

封包解析提供的是獨立的 CoE / SDO Record

單一 Datagram 無法直接表示一次完整的 SDO object access，
Request 與 Response 必須跨封包建立關聯。

- Request / Response 分散
- Index / SubIndex 資訊分散
- Result / Abort 狀態需要整合

02 GOAL

重建完整的 SDO Transaction

將 Mailbox / CoE / SDO 欄位重新組合，
建立可供後續 PDO Mapping 分析使用的 transaction-level model。

- Request ↔ Response
- Index / SubIndex
- Access Direction
- Result / Abort

INPUT
Mailbox / CoE Datagram

PROCESS
Transaction Reconstruction

OUTPUT
SDO Transaction Model

SDO Transaction Reconstruction Flow

PACKET LEVEL → TRANSACTION LEVEL

01	02	03	04	05	06
DATAGRAM	MAILBOX	CoE	SDO	MATCH	OUTPUT
EtherCAT Datagram	Mailbox Detection	CoE Classification	SDO Command Extraction	Request / Response Matching	SDO Transaction

EtherCAT Spec 重點整理

01
KEY CONCEPT

Core concept or mechanism

Summarize the core mechanism, state transition, or protocol rule that defines the behavior described in the specification. Note what the device or analyzer should observe when this rule is applied.

02
IMPORTANT FIELDS

Registers / objects / parameters

Record register addresses, object indices, data types, access rules, and parameter constraints. Highlight which fields are read-only, writable, or dependent on operating state.

03
DATA FLOW

How data moves through the system

Trace how the request, datagram, mapped memory, and application data move through the system. Mark the boundaries where timing, ownership, or synchronization affects the result.

04
ENGINEERING NOTE

Implementation or analyzer relevance

Connect the specification detail to source code, packet analysis, validation steps, or observable system behavior. Record the implementation risk or evidence that should be checked.

EtherCAT 圖解分析

FIGURE / 04

[Diagram Canvas]

Class Diagram / Architecture Diagram / Data Path Figure

REPLACE WITH IMG / SVG / FIGURE / HTML DIAGRAM

FOCUS POINTS

01 **Diagram Purpose**
Explain the main relationship between modules.

02 **Key Components**
Highlight the major classes or architecture blocks.

03 **Reading Order**
Describe how to read the diagram efficiently.

04 **Implementation Relevance**
Connect the diagram to the current development work.

EtherCAT 資料路徑概覽

[FIGURE]

KNOWLEDGE POINTS

01
FMMU
Logical Address
→ ESC local

02
SyncManager
Consistent data
exchange control

03
Process RAM
Stores mapped
process data

04
DATA PATH
Logical Address → FMMU →
SM → RAM → Application